

Coordination Compounds: Formula Sheet

Class 12 Chemistry – Unit 5 Complete Revision Formulas & Key Theories

1 Werner's Theory & Coordination Entity Terms

- Werner's Linkages:

Primary Valence = Oxidation state of central metal (ionisable, satisfied by anions)

Secondary Valence = Coordination Number (non-ionisable, directional, satisfied by ligands)

- Coordination Number (C.N.):

$$\text{C.N.} = \sum (\text{Number of ligands} \times \text{Denticity of ligand})$$

Only sigma (σ) bonds formed between donor atoms and central metal are counted.

- Oxidation State of Central Metal (x):

$$x + \sum (\text{Charge on each ligand}) = \text{Net charge on the coordination entity}$$

2 Magnetic Properties & Valence Bond Theory (VBT)

- 'Spin-Only' Magnetic Moment (μ):

$$\mu = \sqrt{n(n+2)} \text{ BM} \quad \left(1 \text{ BM} = \frac{eh}{4\pi m_e} = 9.27 \times 10^{-24} \text{ J} \cdot \text{T}^{-1} \right)$$

Where n is the number of unpaired electrons.

- Hybridisation Geometries in Coordination Entities:

C.N. 4 : $sp^3 \Rightarrow$ Tetrahedral (e.g., $[\text{NiCl}_4]^{2-}$, $[\text{Ni}(\text{CO})_4]$)

$dsp^2 \Rightarrow$ Square Planar (e.g., $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{PtCl}_4]^{2-}$)

C.N. 6 : $d^2sp^3 \Rightarrow$ Octahedral / Inner-orbital / Low spin (e.g., $[\text{Co}(\text{NH}_3)_6]^{3+}$)

$sp^3d^2 \Rightarrow$ Octahedral / Outer-orbital / High spin (e.g., $[\text{CoF}_6]^{3-}$)

3 Crystal Field Theory (CFT)

- **Octahedral Crystal Field Splitting (Δ_o):**

$$\text{Energy of } e_g \text{ orbitals } (d_{x^2-y^2}, d_{z^2}) = +\frac{3}{5}\Delta_o = +0.6\Delta_o$$

$$\text{Energy of } t_{2g} \text{ orbitals } (d_{xy}, d_{yz}, d_{xz}) = -\frac{2}{5}\Delta_o = -0.4\Delta_o$$

Crystal Field Stabilization Energy (CFSE) for $t_{2g}^p e_g^q$:

$$\text{CFSE} = (-0.4p + 0.6q)\Delta_o + mP$$

Where P is pairing energy and m is number of extra electron pairs.

- **High Spin vs. Low Spin Criteria in Octahedral Field:**

$$\Delta_o < P \implies \text{Weak field ligand} \rightarrow \text{High spin complex (e.g., } t_{2g}^3 e_g^1 \text{ for } d^4)$$

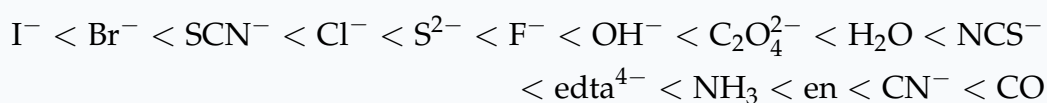
$$\Delta_o > P \implies \text{Strong field ligand} \rightarrow \text{Low spin complex (e.g., } t_{2g}^4 e_g^0 \text{ for } d^4)$$

- **Tetrahedral Crystal Field Splitting (Δ_t):**

$$\Delta_t = \frac{4}{9}\Delta_o$$

e orbitals ($d_{x^2-y^2}, d_{z^2}$) are lowered by $-\frac{3}{5}\Delta_t$; t_2 orbitals (d_{xy}, d_{yz}, d_{xz}) are raised by $+\frac{2}{5}\Delta_t$. ($\Delta_t < P$ always, hence almost exclusively high spin).

- **Spectrochemical Series (Order of Increasing Ligand Field Strength):**



4 Isomerism in Coordination Compounds

- **1. Geometrical Isomerism:**

- **Square Planar** $[Ma_2b_2]$: *cis* and *trans* forms.
- **Square Planar** $[Mabcd]$: 3 geometrical isomers (2 *cis*, 1 *trans*).
- **Octahedral** $[Ma_4b_2]$ / $[M(AA)_2b_2]$: *cis* and *trans* forms.
- **Octahedral** $[Ma_3b_3]$: *facial* (*fac*) and *meridional* (*mer*) forms.

- **2. Optical Isomerism:**

- Non-superimposable mirror images (*dextro* and *laevo*).
- Exhibited by $[M(AA)_3]^{n\pm}$ and *cis*- $[M(AA)_2b_2]^{n\pm}$ (the *trans* isomer is achiral/optically inactive due to a plane of symmetry).

- **3. Structural Isomerism Types:**

- **Linkage Isomerism:** Involves ambidentate ligands (e.g., $-\text{NO}_2^-$ vs. $-\text{ONO}^-$; $-\text{SCN}^-$ vs. $-\text{NCS}^-$).
- **Coordination Isomerism:** Interchange of ligands between complex cation and complex anion (e.g., $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ and $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$).
- **Ionisation Isomerism:** Counter ion displaces a coordinated ligand (e.g., $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$).
- **Solvate / Hydrate Isomerism:** Difference in whether water acts as a ligand or crystal water (e.g., $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ vs. $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$).